MATH 101

# Chapter 4 Homework

Linear functions

**Section 4.1** | Estimated time: about 1 hour

**Complete all problems and show your work on separate paper.** Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Give exact answers. Use a basic four-function calculator only if needed; show algebraic reasoning.

## Recognizing and describing linear functions

**1.** For each table, decide whether a linear function fits all points. Justify using rates of change. If linear, find its rule.

| x | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| A(x) | 2 | 5 | 8 | 11 |
| B(x) | 2 | 4 | 8 | 16 |

**2.** State the slope and y-intercept; decide whether the function increases or decreases. Find f(-2) and solve f(x) = 1.

$$f(x) = -\frac{3}{2}x + 4$$

**3.** Find the equation of the line through the two points. Use function notation.

$$(-2, 5), \quad (4, -1)$$

**4.** Find the slope and y-intercept, then sketch the graph.

$$3x + 2y = 8$$

## Connecting equations, graphs, and rates

**5.** Sketch both equations. Identify which one defines y as a function of x, and explain.

$$x = -2, \qquad y = 3$$

**6.** Find equations of the lines through (2, -1) that are (a) parallel and (b) perpendicular to the given line.

$$y = \frac{2}{3}x + 4$$

**7.** A tank contains 180 liters and loses 6 liters per minute at a constant rate. Write V(t), interpret its slope and intercept with units, and give a physically meaningful domain.

**8.** For the tank in Problem 7, find V(8) and when the volume reaches 60 liters. Explain the difference between these two questions.

## Intersections and reasoning

**9.** Find the intersection algebraically and confirm it by sketching both lines.

$f(x) = 2x + 1, \qquad g(x) = -x + 7$

**10.** Find the input interval on which f(x) is greater than g(x), using the functions in Problem 9. Explain what this means on the graph.

**11.** A line has slope -2 and passes through (3, 5). A student writes y = -2x + 5. Explain the error and give the correct equation.

**12.** Find the inverse of f(x) = 4x - 6. State its slope. Explain why the two graphs are reflections across y = x.